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The Magic of iCLIP 3D PRINTING

THE NEWEST WAY TO 3D PRINT ANYTHING

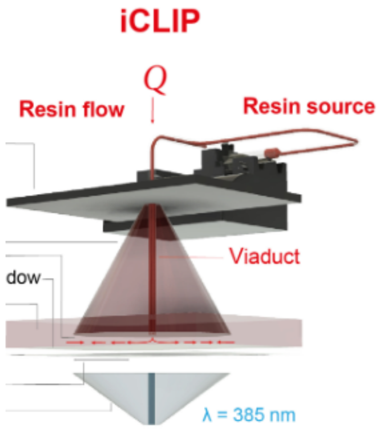
A search for the magic wand

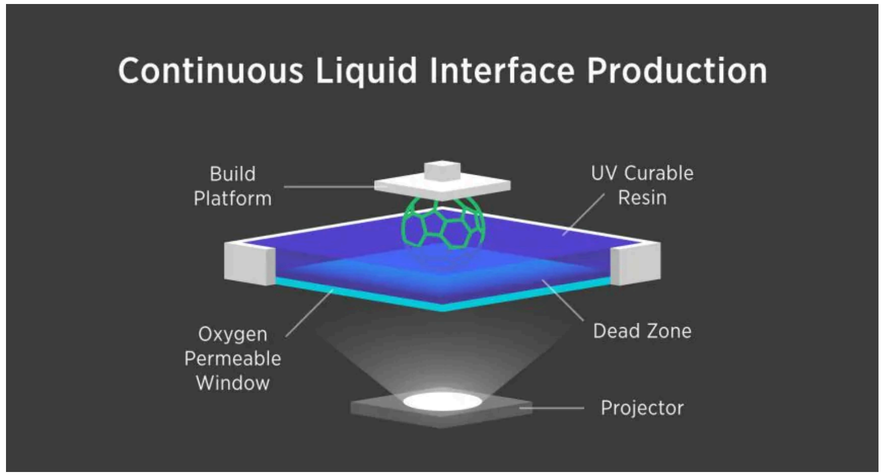
As kids, we all dream of magic wands that can create our wildest fantasies in the blink of an eye. 3D printing has allowed these childhood fantasies of creation to come to life, and the idea of a magic wand is becoming non-fiction through every advancement. iCLIP just might be the closest thing we currently have to a magic wand. Its ability to form complex geometry rapidly out of seemingly nothing is nothing short of magical. Its complex laser system is straight from a sci-fi movie, while being entirely grounded in reality. This fantastical sounding technology isn't the result of pure imagination, but instead years of engineering development.

CLIP - and its shortcomings

As early as 2015, a group of researchers headed by chemist Joseph DeSimone introduced the world to a 3D printing methodology called continuous liquid interface production, or CLIP for short. Imagine a platform slowly rising upwards out of a vat of liquid, forming complex parts. A projector, much like those seen in classrooms, is used to harden the liquid resin layer by layer. Instead of a traditional projector, this technology uses a much more detailed, UV based projecting system. A thin layer of oxygen is kept at the bottom of this liquid pool, which allows the part to slowly rise out of the vat. This layer is known as the “dead zone”, where no resin is able to enter.

For years, CLIP was one of the fastest 3-D printing technologies available, but wasn't widely adopted. This is primarily because of the common defects in printed parts, as getting the resin to flow over the “dead zone”, proved to be difficult with the current setup. For more consistent printing, speeds had to be reduced significantly, making the technology less attractive. If this technology would ever make it into the mainstream additive manufacturing space, it would need a significant change...





The iClP Reimagined

In September 2022, lead researcher Gabriel Lipkowitz and his team published an article in Science Advances looking to solve this exact problem with the technology. By injecting the resin, much like how a syringe works, Lipkowitz added injection to the pre-existing CLIP technology, putting the "i" in iCLIP. Instead of passively allowing the resin to flow, and hope that it does, this new addition forced resin into the proper places, and at the exact right time.

Veins in a 3D print?

With iCLIP, pumps mounted on the rising platform inject resin through "viaducts", what the researchers called the mini guiding channels. Fascinatingly, these viaducts are printed directly into the object as it is being made. These channels act like veins, allowing the blood (resin) of the system to be pumped throughout the body (part) where it is needed. With this addition, the speed at which parts could be produced increased dramatically, on average by six times. These viaducts are integral to the function of the print, just like viens in the body.

This new injection system not only brings a faster, more reliable printing operation, but also unlocks multiple other avenues of expansion. The primary extra benefit of these "viaducts" is the ability to print multiple types of materials at once in the same part. Researchers demonstrated this by printing models of various colors, such as the Independence Hall in red, white and blue. Beyond colors, multimaterial printing unlocks things such as printing, rigid and flexible materials in the same part, allowing for things like hinges built in.

Beyond the speed

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The researchers furthered this technology by injecting resin with carbon nanotubes, something no other 3-D printing technology can reliably manufacture with. These nanotubes can be printed with this technology due to the higher pressure created by the injection system. Nanotubes act as a much more rigid structure for the 3D printed parts. These nanotubes act as the structure of the print - they act as the bones. These "bones" provide great strength, unlocking many possibilities in terms of advanced 3d prints.

How this helps us

The researchers furthered this technology by injecting resin with carbon nanotubes, something no other 3-D printing technology can reliably manufacture with. These nanotubes can be printed with this technology due to the higher pressure created by the injection system. Overall, this system acts like the body, with viaducts as veins and nanotubes as bones. These additions draw us closer to a future where "magic wands" are more real than we might have thought!